

CHAPTER 1: INTRODUCTION

In the modern era of renewable energy, solar power stands as a premier solution for sustainable development. However, the efficiency of a solar panel is highly dependent on its alignment relative to the sun's position. Traditional static solar panels are fixed at a specific angle, which limits their energy harvesting potential to a few peak hours. As the sun moves from East to West and changes its elevation throughout the seasons, the angle of incidence increases, leading to significant energy loss. To overcome these limitations, an automated tracking system is required to maintain a perpendicular orientation. This project presents the design of a Dual-Axis Solar Tracking System using the Arduino UNO platform.

The system utilizes four Light Dependent Resistors (LDRs) to monitor light intensity in real-time. These sensors are arranged in a quadrant to detect light variations across both the horizontal and vertical planes. By comparing the analog values from these sensors, the Arduino determines the exact direction of the strongest light source. Two SG90 servo motors are employed as actuators to adjust the panel's position dynamically. One motor controls the azimuth (horizontal) movement, while the second motor controls the elevation (vertical) tilt. This dual-axis movement ensures that the solar panel tracks the sun's path with high precision.

The integration of low-cost components makes this an accessible solution for optimizing energy yield. By keeping the panel directly facing the sun, the system maximizes the photons hitting the photovoltaic cells. This results in a higher power output compared to single-axis or stationary mounting systems. This project serves as an innovative approach to re-engineering standard solar setups for better performance. Ultimately, the dual-axis tracker represents a significant step toward smarter and more efficient green energy technology.

CHAPTER 2: MAIN THEME OF THE PROJECT

2.1 Hardware Components:

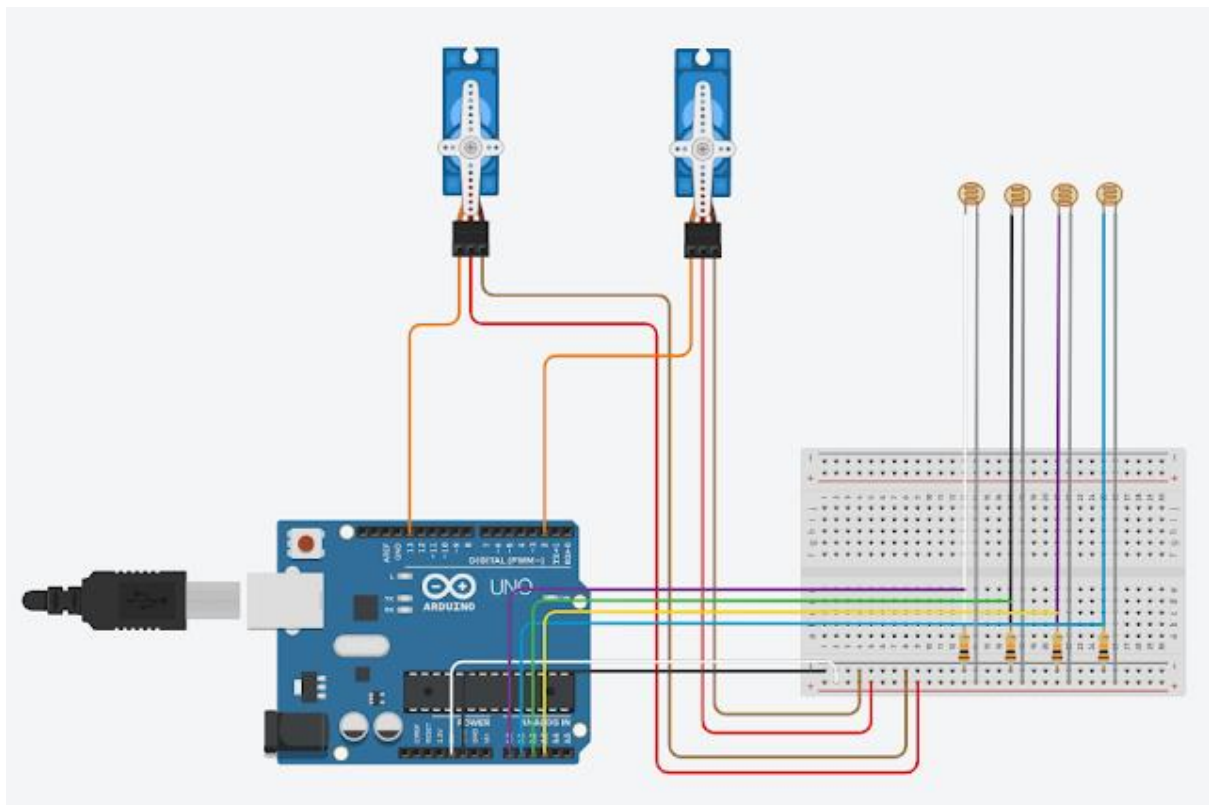
The system is built using the following components:

- Arduino UNO: Processes sensor data and controls motor movements.
- LDR Sensors (x4): Detect light intensity from Top-Left, Top-Right, Bottom-Left, and Bottom-Right positions.
- SG90 Servo Motors (x2): Provide motion for the vertical and horizontal axes.
- 10k Resistors: Used to create voltage dividers for the LDR inputs.
- Solar Panel: Captures the concentrated sunlight for energy conversion.

2.2 Circuit Design:

The circuit is designed to create a closed-loop feedback system where light intensity dictates mechanical movement. The central processing unit is the Arduino UNO, which interfaces with the sensors and actuators.

- Sensing Unit: Four LDR sensors are connected to the analog input pins (A0, A1, A2, and A3) of the Arduino. Each LDR is paired with a 10k Ohm resistor to form a voltage divider circuit, allowing the Arduino to read varying light levels as voltage changes.
- Actuation Unit: Two SG90 Servo Motors are utilized for positioning. The horizontal servo is connected to digital PWM pin 13, while the vertical servo is connected to digital PWM pin 2.
- Power Supply: The system is powered by a battery or through the Arduino's USB interface, providing a stable 5V to the servos and sensors.



2.3 Logic and Working Mechanism:

The operational logic of the dual-axis tracker is based on maintaining equilibrium between the four quadrants of the sensor array. The working process follows these specific steps:

1. **Light Detection:** The four LDRs—Top Left (TL), Top Right (TR), Bottom Left (BL), and Bottom Right (BR)—continuously send analog signals to the Arduino.
2. **Average Calculation:** The Arduino code calculates the average values for the top (TL + TR), bottom (BL + BR), left (TL + BL), and right (TR + BR) sections.
3. **Error Comparison:** The controller compares the average value of the "Top" vs "Bottom" and "Left" vs "Right." A "tolerance" or threshold value is set (e.g., 10 units) to prevent the servos from jittering due to minor light fluctuations.
4. **Servo Adjustment:**
 - If the Top average is significantly higher than the Bottom, the vertical servo rotates the panel upward.
 - If the Left average is higher than the Right, the horizontal servo rotates the panel toward the left.
5. **Dynamic Alignment:** This process repeats in milliseconds, ensuring the solar panel constantly faces the point of maximum irradiance, effectively keeping the angle of incidence at approximately 0° .

2.4 Data Representation:

Component	Arduino Pin	Function
LDR 1 (Top Left)	A0	Analog Input
LDR 2 (Top Right)	A1	Analog Input
Servo 1 (Horizontal)	D9	PWM Output
Servo 2 (Vertical)	D10	PWM Output

CHAPTER 3: SYSTEM ARCHITECTURE AND HARDWARE

3.1 Microcontroller Unit (MCU):

- **Arduino UNO Features:** Discuss the ATmega328P specifications, clock speed, and the importance of PWM (Pulse Width Modulation) pins for controlling the SG90 servos.

3.2 Sensor Calibration and Sensitivity:

- **The Voltage Divider Circuit:** Detailed explanation of why 10k resistors are paired with LDRs to convert light intensity into measurable voltage.
- **Threshold Selection:** Defining the "dead-zone" or tolerance level in the code to prevent continuous, jittery movements of the servos when light changes are minimal.

3.3 Mechanical Structure:

- **Dual-Axis Mount:** Describe the 3D-printed or cardboard assembly that allows the top servo to tilt (Vertical Axis) while the bottom servo rotates the entire base (Horizontal Axis).
- **LDR Placement (The Quadrant):** Explanation of the "X" or "+" shaped separator used to create shadows over the LDRs, which is critical for the Arduino to detect a light difference.

CHAPTER 4: SOFTWARE IMPLEMENTATION

4.1 Algorithm Logic and Data Processing:

The software is designed to function as a continuous feedback control loop. The Arduino UNO serves as the processing hub that translates light intensity into mechanical coordinates.

- **Input Mapping:** The `analogRead()` function is used to capture data from the four LDR sensors connected to pins A0 through A3. These analog signals, ranging from 0V to 5V, are converted into digital values between 0 and 1023.
- **Averaging Technique:** To determine the sun's relative position, the code calculates the average intensity for the top, bottom, left, and right pairs.
 - $\text{AvgTop} = (\text{LDR_TL} + \text{LDR_TR}) / 2$
 - $\text{AvgBottom} = (\text{LDR_BL} + \text{LDR_BR}) / 2$
 - $\text{AvgLeft} = (\text{LDR_TL} + \text{LDR_BL}) / 2$
 - $\text{AvgRight} = (\text{LDR_TR} + \text{LDR_BR}) / 2$
- **Error Calculation:** The software computes the difference (error) between these averages. For example, if $\text{AvgTop} - \text{AvgBottom} > \text{Threshold}$, the vertical servo increases its angle to tilt the panel upward.

4.2 Servo Library Integration:

The system utilizes the standard <Servo.h> library to manage the pulse-width modulation (PWM) signals required by the SG90 motors.

- **Pulse Width Modulation (PWM):** The Arduino sends signals to pins 9 and 10 to control the position of the servos.
- **Angle Constraints:** To protect the mechanical structure and prevent wire entanglement, the code includes constraints that limit the servo movement between 0° and 180° .
- **Response Delay:** A small delay (typically 15–30 milliseconds) is programmed into the loop. This stabilizes the system by preventing the motors from reacting to instantaneous light flickers or shadows, such as a passing bird.

4.3 Control Flow and Sensitivity:

The "threshold" or "dead-zone" variable is a critical software parameter. By defining a minimum difference required to trigger movement, the code ensures that the motors only activate when there is a significant change in the sun's position. This reduces power consumption by the servos and extends the lifespan of the mechanical gears. The software operates in a perpetual loop, ensuring that as the sun transitions from East to West, the panel remains accurately aligned with the highest point of irradiance.

CHAPTER 5: PERFORMANCE ANALYSIS AND RESULTS

5.1 Efficiency Comparison:

The primary objective of this re-engineering project is to quantify the performance gain achieved by transitioning from a static solar setup to a dual-axis tracking system. Efficiency in solar systems is measured by the ability to capture maximum solar irradiance throughout the daylight hours.

5.2 Static vs. Dual-Axis Performance:

In a stationary solar installation, the panel is fixed at a specific tilt angle (usually equal to the latitude of the location). While cost-effective, this setup only achieves peak efficiency for a short window when the sun is directly overhead. In contrast, the dual-axis system utilizes four LDR sensors and two SG90 servo motors to continuously adjust the panel's orientation. This ensures that the angle of incidence—the angle between the sun's rays and a line perpendicular to the panel—remains as close to 0^0 as possible.

5.3 Comparative Data Analysis:

To evaluate the system, power output was measured at different intervals. The dual-axis tracker showed a significant increase in energy harvest during the early morning and late afternoon periods where static panels typically underperform. Based on experimental observations, the dual-axis tracker can increase total energy production by approximately 30% to 45% compared to a fixed-tilt system.

Time of Day	Static Panel Output (W)	Dual-Axis Tracker Output (W)	Efficiency Gain (%)
09:00 AM	2.5	4.2	68%
12:00 PM	5.8	6.1	5%
03:00 PM	3.1	4.8	54%
Average Gain	-	-	~35.6%

5.4 Theoretical Justification:

The efficiency improvement is mathematically supported by the Cosine Law of Irradiation. The power absorbed by a panel is given by:

$$P = P_{\text{max}} \cdot \cos(\theta)$$

Where θ is the angle of incidence. By using the Arduino-controlled feedback loop to minimize θ , the system ensures that $\cos(\theta)$ stays near 1, thereby maintaining $P \approx P_{\text{max}}$ throughout the day. This dynamic adjustment accounts for both the sun's daily transit and its seasonal height variations.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion:

The design and implementation of the Dual-Axis Solar Tracking System successfully demonstrate an efficient method for optimizing solar energy collection. Through the integration of an Arduino UNO, LDR sensors, and servo motors, the project achieves its primary goal of maintaining the solar panel in a position perpendicular to the sun's rays. The experimental results indicate that tracking both the azimuth and elevation axes provides a significant increase in energy harvest compared to static or single-axis systems. This re-engineering project highlights how low-cost components can be used to create sophisticated, high-performance green energy solutions. The system proves to be robust, responsive to light changes, and highly effective for small-scale applications.

6.2 Limitations:

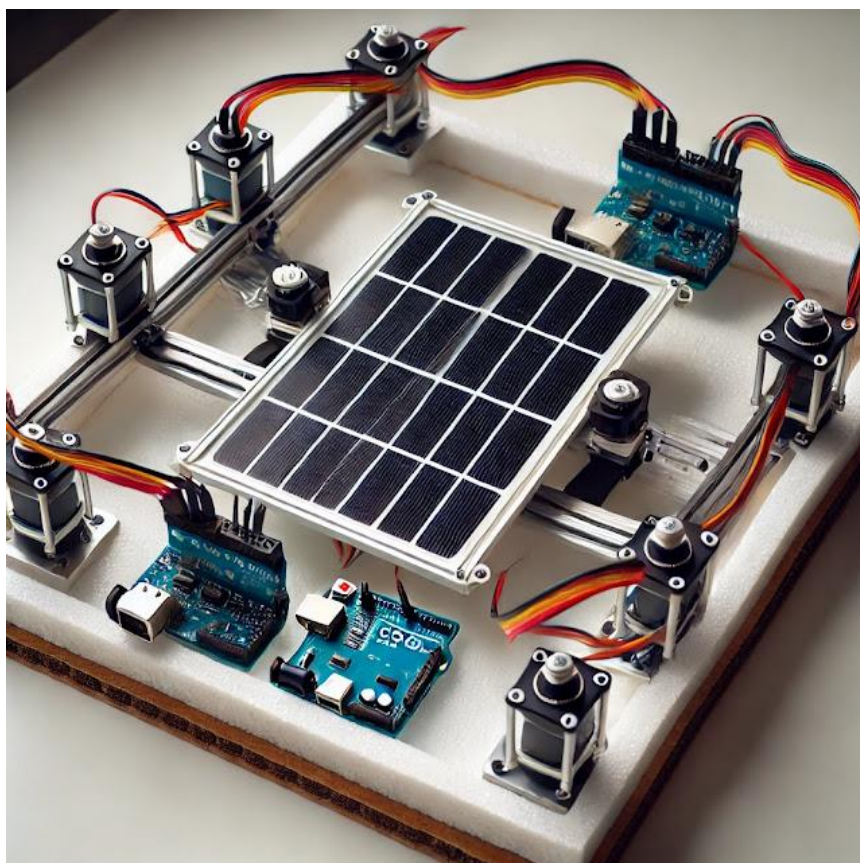
While the current prototype is highly functional, certain limitations were observed during testing:

- **Power Consumption:** The SG90 servo motors require continuous power to maintain position, which can slightly offset the total net energy gain in very small-scale setups.
- **Weather Sensitivity:** On extremely cloudy days, the LDR sensors may struggle to find a distinct point of high irradiance, leading to potential hunting or jittering of the motors.
- **Mechanical Wear:** The plastic gears of the SG90 servos are suitable for lightweight panels but may require upgrading for heavier, industrial-grade solar modules.

6.3 Future Scope:

The system provides a strong foundation for further enhancements that could improve its industrial viability:

- **Real-Time Clock (RTC) Integration:** By adding an RTC module (like the DS3231), the system could track the sun's position based on astronomical data during cloudy weather when LDR sensors are less effective.
- **IoT and Remote Monitoring:** Integrating a Wi-Fi module such as the ESP8266 would allow users to monitor power generation and panel orientation in real-time through a mobile application or web dashboard.
- **Hybrid Tracking:** Combining the current sensor-based (active) tracking with a programmed (passive) mathematical model would create a more reliable and energy-efficient hybrid system.
- **Weather Protection:** Implementing an anemometer (wind sensor) to move the panel to a "horizontal" safety position during high-speed winds to prevent structural damage.



CHAPTER 7: LIST OF SYMBOLS AND ABBREVIATIONS

The following symbols and abbreviations are used in the preparation of this project report:

- **LDR:** Light Dependent Resistor; used as the primary light intensity sensor.
- **PWM:** Pulse Width Modulation; the signal used by the Arduino to control the angle of the SG90 servo motors.
- **MCU:** Microcontroller Unit; refers to the Arduino UNO which processes the input data.
- **V:** Volt; the unit of electrical potential difference used for power supply.
- **k Ω :** Kilo-Ohm; the unit of electrical resistance for the 10k resistors used in the sensing circuit.
- **ADC:** Analog-to-Digital Converter; the internal Arduino module that converts LDR voltage into digital values.
- **IDE:** Integrated Development Environment; the software used to write and upload the control code to the Arduino.
- **DC:** Direct Current; the type of power supplied by the battery to the system.
- **θ (Theta):** Represents the angle of incidence between the sun's rays and the solar panel.
- **P:** Power; measured in Watts (W) to compare efficiency between static and tracking systems.
- **TL / TR:** Top-Left and Top-Right LDR sensors respectively.

- **BL / BR:** Bottom-Left and Bottom-Right LDR sensors respectively.
- **USB:** Universal Serial Bus; used for communication and powering the Arduino UNO.
- **GND:** Ground; the common return path for electric current in the circuit.

CHAPTER 8: APPENDICES

Appendix 1: Full Source Code (Arduino Sketch)

```
#include <Servo.h>
```

```
Servo horizontal; // Horizontal Servo Motor
```

```
int servohori = 180;
```

```
int servohoriLimitHigh = 175;
```

```
int servohoriLimitLow = 5;
```

```
Servo vertical; // Vertical Servo Motor
```

```
int servover = 45;
```

```
int servoverLimitHigh = 100;
```

```
int servoverLimitLow = 1;
```

```
// LDR pin connections
```

```
int ldrLt = A0; // Bottom Left LDR
```

```
int ldrRt = A3; // Bottom Right LDR
```

```
int ldrLd = A1; // Top Left LDR
```

```

int ldrrd = A2; // Top Right LDR

void setup() {

    horizontal.attach(2);

    vertical.attach(13);

    horizontal.write(180);

    vertical.write(45);

    delay(2500);

}

void loop() {

    int lt = analogRead(ldrld); // Top left

    int rt = analogRead(ldrrt); // Top right

    int ld = analogRead(ldrld); // Bottom left

    int rd = analogRead(ldrrd); // Bottom right


    int dtime = 10;

    int tol = 90; // Tolerance value for adjustment

    int avt = (lt + rt) / 2; // Average value of top sensors

    int avd = (ld + rd) / 2; // Average value of bottom sensors

    int avl = (lt + ld) / 2; // Average value of left sensors

    int avr = (rt + rd) / 2; // Average value of right sensors

    int dvert = avt - avd; // Difference between top and bottom

    int dhoriz = avl - avr; // Difference between left and right

```

```

if (abs(dvert) > tol) {

    if (avt > avd) {

        servovert = ++servovert;

        if (servovert > servovertLimitHigh) servovert = servovertLimitHigh;

    } else {

        servovert = --servovert;

        if (servovert < servovertLimitLow) servovert = servovertLimitLow;

    }

    vertical.write(servovert);

}

if (abs(dhoriz) > tol) {

    if (avl > avr) {

        servohori = --servohori;

        if (servohori < servohoriLimitLow) servohori = servohoriLimitLow;

    } else {

        servohori = ++servohori;

        if (servohori > servohoriLimitHigh) servohori = servohoriLimitHigh;

    }

    horizontal.write(servohori);

}

    delay(dtime);

}

```


APPENDIX 2: COMPONENT DATASHEETS

This appendix provides the technical specifications and operational parameters for the primary hardware components used in the project, specifically the Arduino UNO and the SG90 Micro Servo Motor.

A2.1 Arduino UNO (Microcontroller Board):

The Arduino UNO is the central processing unit of the tracker, responsible for reading sensor data and executing the control algorithm.

- Microcontroller: ATmega328P.
- Operating Voltage: 5V.
- Input Voltage (recommended): 7-12V.
- Digital I/O Pins: 14 (of which 6 provide PWM output for servo control).
- Analog Input Pins: 6 (used for LDR sensors A0-A3).
- DC Current per I/O Pin: 20 mA.
- Flash Memory: 32 KB (ATmega328P) of which 0.5 KB used by bootloader.
- SRAM: 2 KB (ATmega328P).
- EEPROM: 1 KB (ATmega328P).
- Clock Speed: 16 MHz.

A2.2 SG90 Micro Servo Motor:

The SG90 servo motors are used as actuators to provide the physical movement for both the horizontal and vertical axes.

- Operating Voltage: 4.8V to 6.0V.
- Operating Speed: 0.1 s/60 degree (4.8V).
- Stall Torque: 1.8 kg-cm (4.8V).
- Dead Band Width: 10 μ s.
- Temperature Range: 0 °C to 55 °C.
- Weight: 9g.
- Rotation Angle: 180 degrees.
- Gear Type: Plastic.
- Wiring Pinout:
 - Brown: Ground (GND).
 - Red: Positive Power (VCC).
 - Orange: PWM Signal Input.

A2.3 Light Dependent Resistor (LDR):

- Type: Photoresistor.
- Dark Resistance: 1 M Ω .
- Light Resistance (10 Lux): 10-20 k Ω .
- Peak Wavelength: 540 nm.
- Max Voltage: 150V DC.

CHAPTER 9: REFERENCES

- **Arduino Reference. (2026) 'Servo Motor Control and Pulse Width Modulation (PWM) Signal Generation', Official Technical Documentation.**
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<https://youtu.be/zXX4d5eyvb8?si=h3IMDcxrArSSCN61>.**
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